Supplementary material for "A continental-scale analysis of genomic variation in a songbird"

Table S1 Sampling details for the 28 European great tit populations.

Population	Region	N	Latitude	Longitude
Loch Lomond	Scotland	21	56.108	-4.562
Cambridge	England	28	52.198	0.152
Oxford	England	48	51.776	-1.303
Font Roja	Spain	19	38.664	-0.544
Mariola	Spain	21	37.717	-0.658
La Rouviere	France	19	43.930	4.235
Montpellier	France	36	43.098	3.569
Antwerp	Belgium	15	51.226	4.390
Westerheide	Netherlands	50	52.250	5.190
Vlieland	Netherlands (Vlieland)	15	53.238	4.932
Zurich	Switzerland	22	47.383	8.523
Radolfzell	Germany	18	47.759	8.957
Seewisen	Germany	30	47.978	11.233
Pirio Muro	Corsica	27	42.110	9.104
Sardinia	Sardinia	9	40.056	9.025
Italy	Italy	13	42.386	13.354
Vienna	Austria	27	48.239	16.347
Velky Kosir	Czech Republic	22	49.547	17.062
Pilis Mountains	Hungary	28	47.733	18.917
Gotland	Sweden	32	57.504	18.466
Harjavalta	Finland	44	61.314	22.139
Oulu	Finland	29	65.013	25.468
Tartu	Estonia	32	58.375	26.739
Crete	Crete	4	35.183	24.960
Bulgaria	Balkans	3	42.852	24.986
Romania	Balkans	7	46.497	24.582
Turkey	Turkey	11	39.464	34.797
Zvenigorod	Russia	17	55.738	36.857

Table S2 Details of outlier F_{ST} regions in European great tit populations. For each region, F_{ST} was calculated for each population versus Turkey. The "number of outliers represents" the number of *populations* in which each region was detected as an outlier.

Chromosome	Position..MB.	Number.of.outliers	Outlier.comparisons	Genes
2	81 5	12	Austria, Belgium, Czech Republic, Estonia, Finland, France Germany, Hungary, Netherlands, Scotland, Sweden, Switzerland	PADP7, SRDSA1, NSUN2, UBE2QL, MED10
6	18	10	Belgium, Corsica, England, France, Germany, Hungary, Italy Netherlands, Russia, Sweden	RASGEF1A, RET, CSGALNACT2
3	28 5	2	England, Spain	PPP1CB, CALM2, TTC7A, MCFD2
1A	70 5	2	Czech Republic, Russia	CDKN1B
4A	12	2	England, Scotland	COL4A5
6	8	2	Finland, Scotland	SLC18A3, RGR, LRI1, LRI2, CDHR1, GHITM
3	28	1	Spain	MRPL33, RBKS, BRE, FOSL2
1A	48 5	1	Finland	CBX6, CBX7, DNAL4, SUN2, NPTXR, JOSD1, TOMM22, KDELR3, DMC1, GTPBP1, DDX17, CBY1
				KCNJ4, THEM184B, CSNK1E, PLA2G6, MAFF, BAIAP2L2, SLC16A8, PICK1, SOX10, MICALL1 EIF3L, ANKRDS4, GCAT, POLR2F
6	7 5	1	Scotland	BMPR1A, SNGC, MMRN2, GLUD1, FAM35A, GPRIN2, PARG, NCOA4, OGDHL, CHAT
8	7 5	1	Vlieland	-
8	8	1	Belgium	-

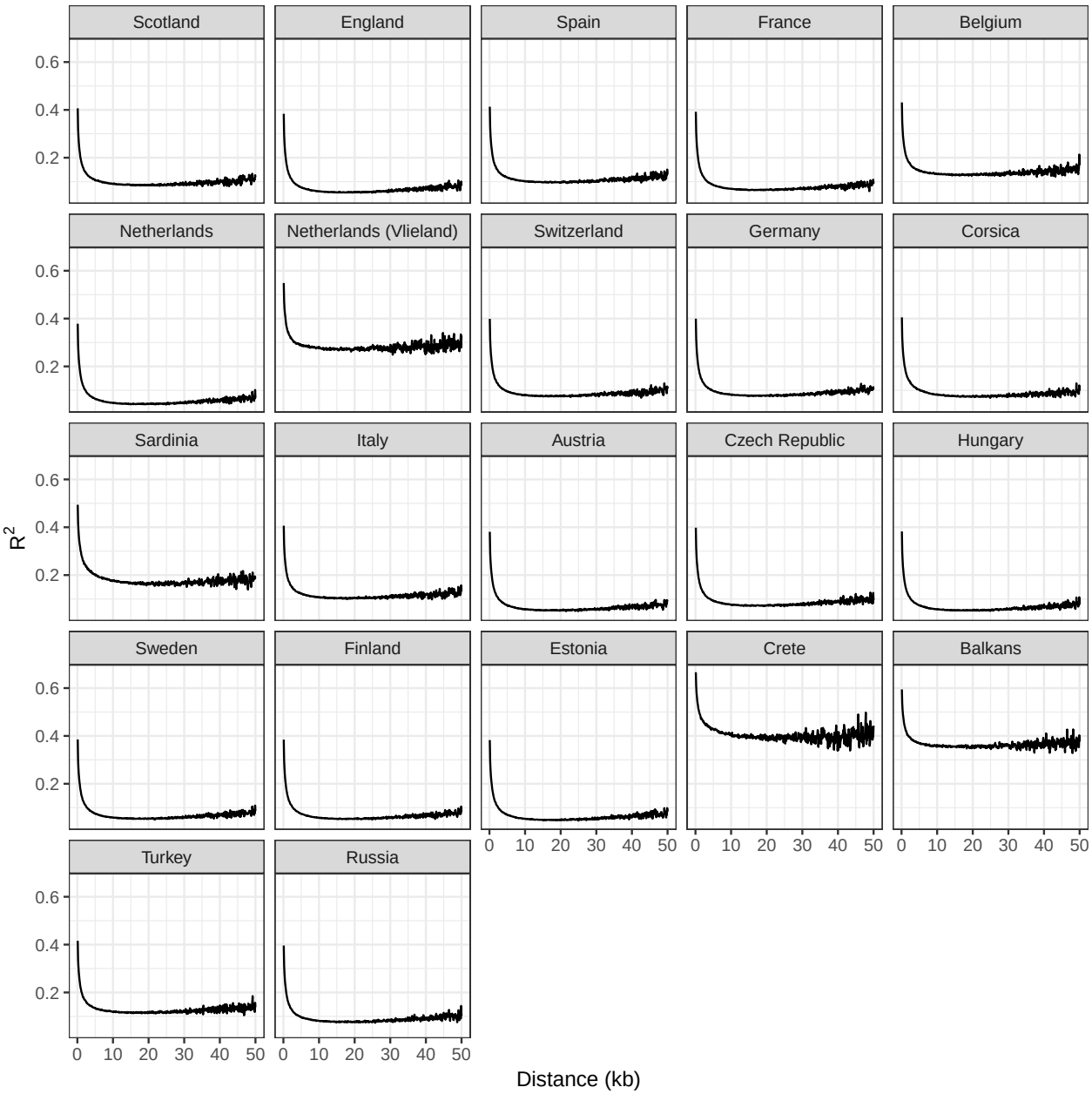

Figure S1 Linkage disequilibrium in European great tit populations. Lines are means of R^2 values for all pairs of markers within 100bp distance bins.

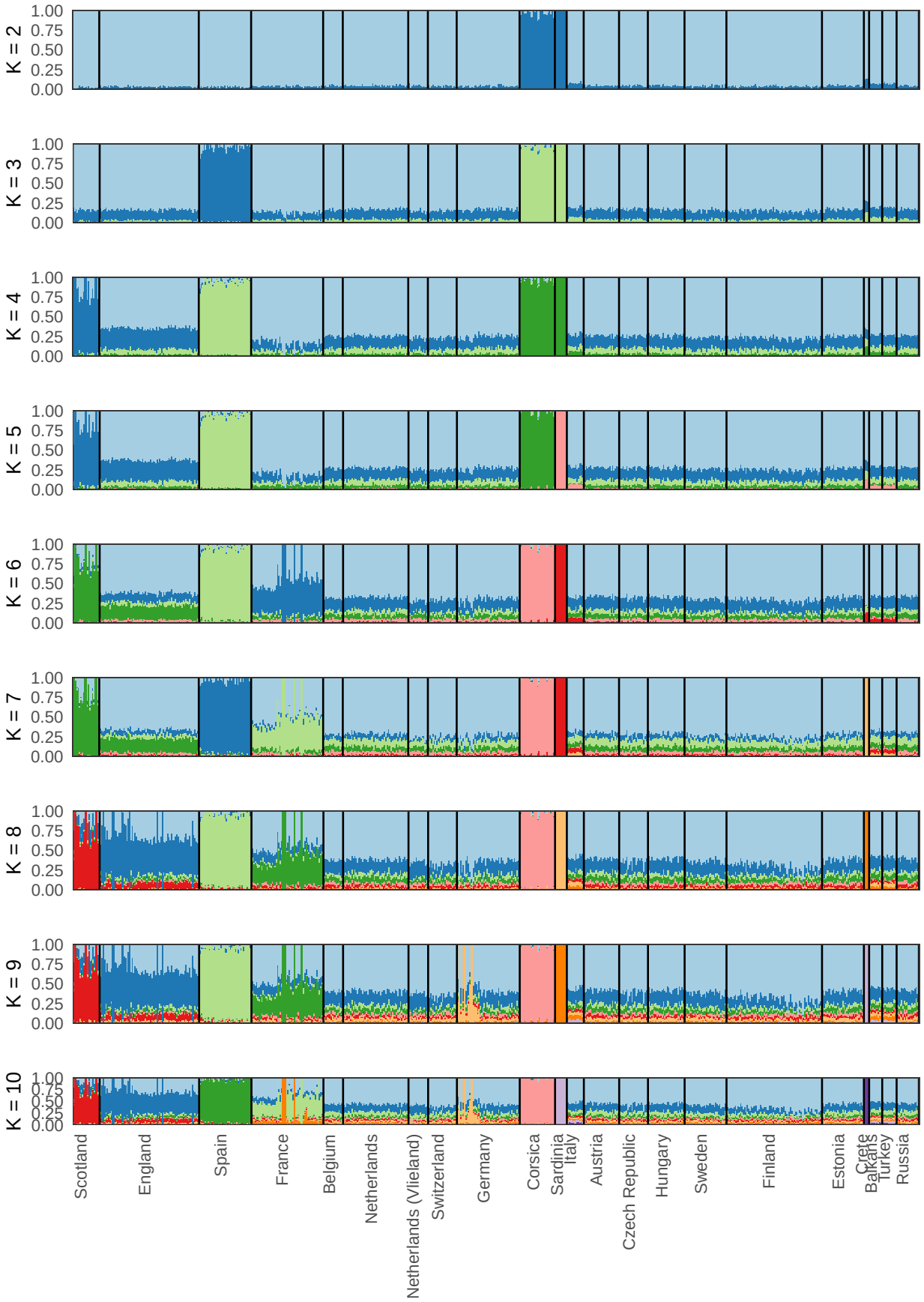
K = 2
K = 3
K = 4
K = 5
K = 6
K = 7
K = 8
K = 9
K = 10
1.00
0.75
0.50
0.25
0.00
Scotland
England
Spain
France
Belgium
Netherlands
Netherlands (Vlieland)
Switzerland
Germany
Corsica
Sardinia
Italy
Austria
Czech Republic
Hungary
Sweden
Finland
Estonia
Crete
Balkans
Turkey
Russia

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Figure S2 Admixture analyses at $K = 2$ to 10 in European Great tit populations.

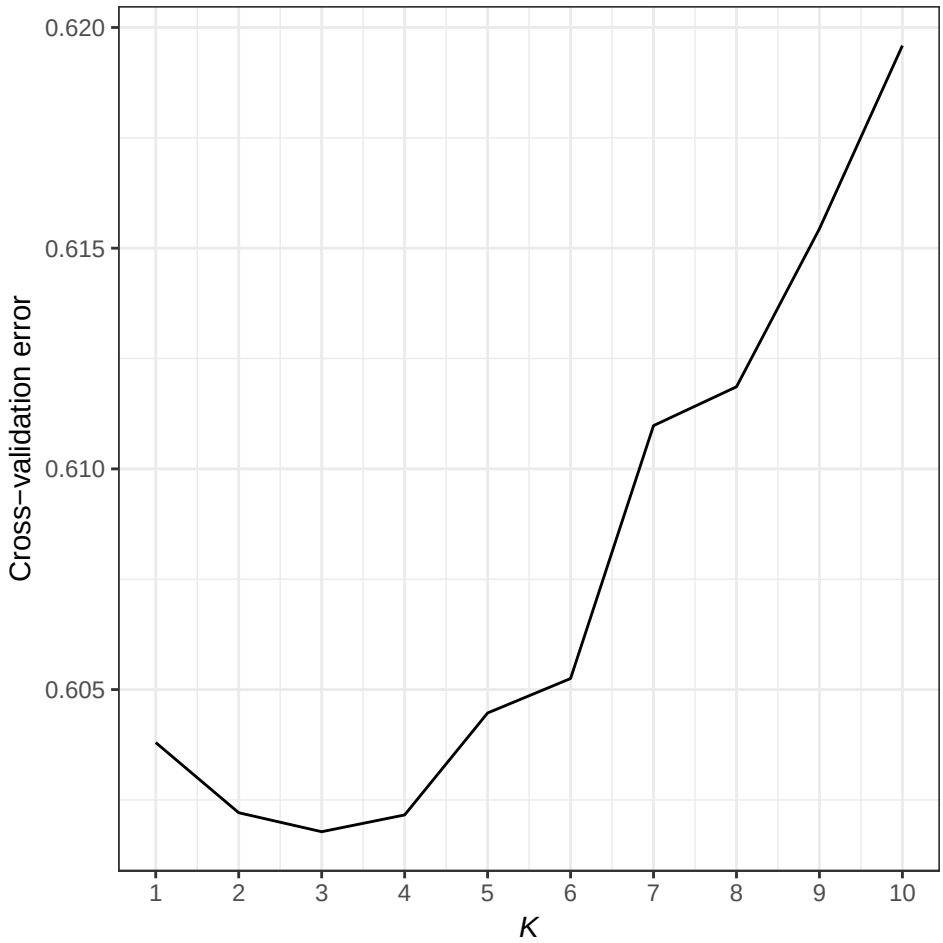

Figure S3 Cross-validation error from admixture analyses at $K = 1$ to 10.

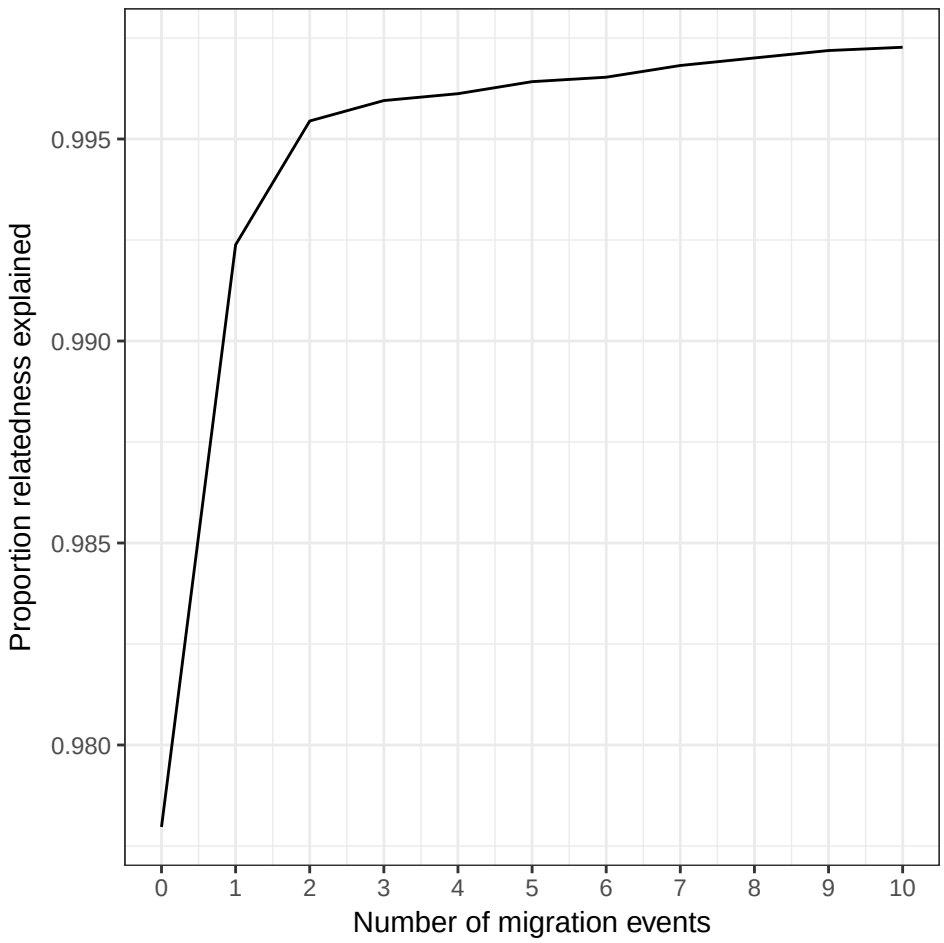

Figure S4 Proportion of relatedness explained among populations in relation to the number of migration events allowed in Treemix analyses.

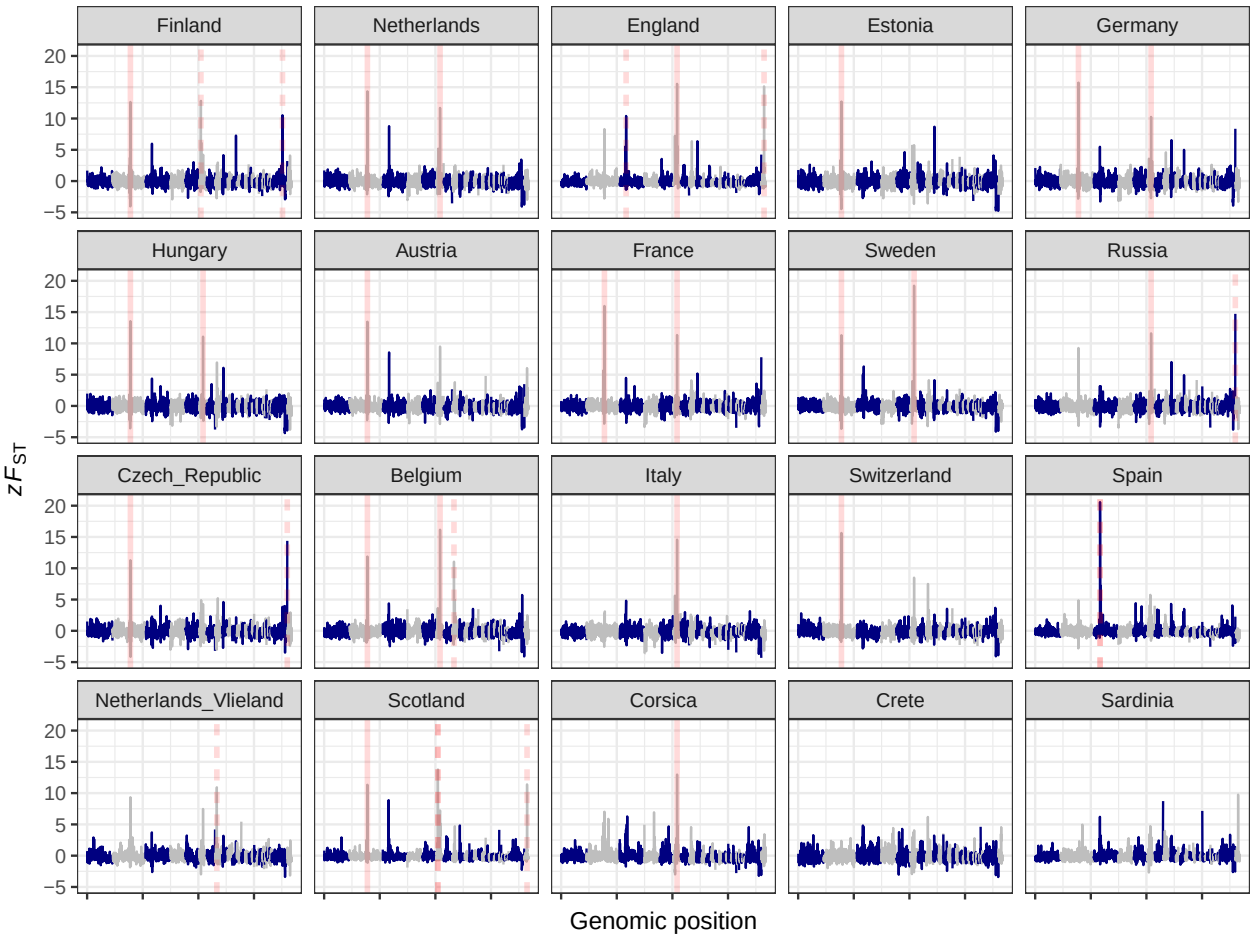

Figure S5 Landscapes of relative genomic differentiation in European great tit populations. zF_{ST} across the genome is averaged in 500kb windows, with each panel displaying a pairwise comparison between with the proposed refugial population in Turkey. Red lines represent F_{ST} outliers (windows with mean $zF_{ST} > 10$) shared across more than two comparisons (solid red lines), or unique to one or two comparisons (dashed red lines).